Practice Problems
Calculus Physics II

Name: _____ **Date:** _____

Chapter 41 Current, Resistance and Ohm's Law

1) An ampere is a unit of electric

(A) current.
B) voltage.
C) resistance.

$I = \frac{dq}{dt}$ $\quad 1A = 1\ C/s$

2) Electrons flow in a wire when there is

A) free electrons in the wire.
(B) an electric field in the wire
C) conduction in the wire.
D) none of the above

$\vec{E}$ create the Force that the charges experience.

3) What current is flowing in a wire if 0.67 C of charge pass a point in the wire in 0.30 s?

$I = \frac{\Delta q}{\Delta t} = \frac{0.67C}{0.3} = 2.23A$

4) When a current flows through a metal wire, the moving charges are

A) only protons.
(B) only electrons.
C) both protons and electrons.
D) positive metal ions.
E) negative metal ions.

5) 5 A of current is following through a 10 Ω resistor, what is the potential difference across the resistor?

$V = IR = (5A)(10\Omega) = 50V$

6) A 4,000 Ω resistor is connected across a 220 V power source. What current will flow through the resistor?

$V = IR \mid I = \frac{V}{R} = 0.055A$

7) A 12 V battery is connected to a light bulb, if 2 A of current flow through the bulb, what is the resistance of the bulb?

$V = IR \mid R = \frac{V}{I} = 6\Omega$

8) A light bulb operating at 110 V draws 1.40 A of current. What is its resistance?

$V = IR \mid R = \frac{V}{I} = 78.6\Omega$

9) Reduce the radius of a wire with a circular cross section by half, what happens to the resistance of the wire?

A) Decreased by a factor of 4.
B) Decreased by a factor of 2.
C) No change in the resistance of the wire.
(D) Increased by a factor of 4.
E) Increased by a factor of 2.

$R = \frac{\rho L}{A} = \frac{\rho L}{(\pi r^2)} = \frac{\rho L}{\pi (\frac{1}{2} r)^2} = 4\left(\frac{\rho L}{\pi r^2}\right)$

10) If two copper wires of the same length have different thickness, then the thicker wire has

A) more resistance.
(B) less resistance.
C) both the same

$R = \frac{\rho L}{A}$ Larger $A \rightarrow$ smaller R

11) A 1.0-mm diameter extension cord is made of metal having a resistivity of $1.68 \times 10^{-8}\ \Omega \cdot m$. When it carries a current of 15 A, what is the potential difference between two points in the cord that are 100 m apart?

$r = 0.5 \times 10^{-3}\ m$

$A = \pi r^2 = 7.85 \times 10^{-7}\ m^2$

$R = \frac{\rho L}{A} = 2.14\ \Omega$

$V = IR$

$V = (15A)(2.14\Omega) = 32V$

12) A certain metal has a resistivity of $1.68 \times 10^{-8}\ \Omega \cdot m$. You have a long spool of wire made from this metal. If this wire has a diameter of 0.15 mm, how long should you cut a segment so its resistance will be 15 Ω?

$r = 0.075\ mm = 0.075 \times 10^{-3}\ m$

$A = \pi r^2 = 1.77 \times 10^{-8}\ m^2$

$R = \frac{\rho L}{A}$ Solve For L $\quad L = \frac{AR}{\rho}$

$L = 15.8\ m$

13) Heat a copper wire and its electric resistance

A) decreases.
B) remains unchanged.
(C) increases.

14) A wire with a cross sectional area of $A = 8 \times 10^{-7}\ m^2$ is made of copper having a resistivity of $1.72 \times 10^{-8}\ \Omega \cdot m$., if the voltage across 500 m of the wire is 120 V,

A) How much current flows when the wire is at 50 °C if it starts from 20 °C? (Assume the Coefficient of Resistivity is $\alpha = 4.0 \times 10^{-3}$ /°C)

$R_{20°C} = \frac{\rho L}{A} = 10.75\Omega$

$R_{50°C} = R_{20°C}(1 + \alpha(\Delta T))$

$R_{50°C} = (10.75\Omega)(1 + (4.0 \times 10^{-3}/°C)(30°C)) = 12.04\Omega$

B) What is the difference in current flow from 20 °C to 50 °C?

$I_{20°C} = \frac{120V}{10.75\Omega} = 11.16A$

$I_{50°C} = \frac{120V}{12.04\Omega} = 9.97A$

$\Delta I = 1.2A$

15) A long copper wire at 0° C is hooked up to a 12 V power source and a current of 3.58 A flows through the wire, the wire heats up to 50° C, how much current is flowing through the wire at 50° C, if the power source stays at 12 V? (For copper $\alpha = 4.3 \times 10^{-3}$ /°C)

$V = I_{0°} R_{0°}$

$R_{0°} = \frac{V_{0°}}{I_{0°}} = 3.35\,\Omega$

$R_{50°} = R_{0°}(1 + \alpha(\Delta T))$

$R_{50°} = 4.07\,\Omega$

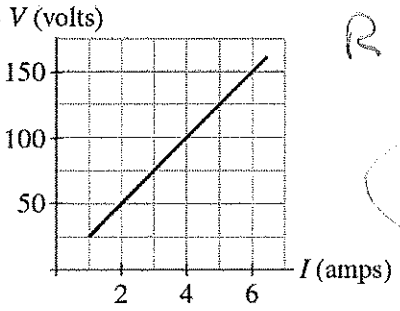
$I_{50°} = \frac{V}{R_{50°}} = \frac{12\,V}{4.07\,\Omega} = 2.95\,A$

16) The graph shown in the figure shows the results of measurements of the dc current through a circuit device for various potential differences across it. Assume that all the numbers shown are accurate to two significant figures. What is the resistance of this device?

A) 100 Ω
B) 150 Ω
C) 50 Ω
D) 25 Ω
E) 75 Ω

$Slope = \frac{\Delta V}{\Delta I}$

$R = Slope$



$R = \frac{V_f - V_i}{I_f - I_i} = \frac{100\,V}{4\,A}$

$R = 25\,\Omega$

17) If a material is non-ohmic, what do we know about the materials resistance?

A) It is constant as the voltage is increased.
B) It does not remain constant as the voltage is increased.
C) The resistance increases with the voltage.

KEY

1) A
2) B
3) 2.23 A
4) B
5) 50 V
6) 0.055 A
7) 6 Ω
8) 78.6 Ω
9) D
10) B
11) 32 V
12) 15.8 m
13) C
14) A) 12.04 Ω B) 1.2 A
15) 2.95 A
16) 25 Ω
17) B